

BrickAtlas: Separating Visual Binding and Compositional Reasoning in Source-Grounded Repair Problems

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Abstract

Recognizing a part and solving a structural problem about it are different abilities. An end-to-end answer can obscure where their composition fails. BrickAtlas evaluates this interface through visually bound minimum terminal repair. A model receives an exact connector graph with unavailable vertices, two named terminals, and an image that determines a third terminal through geometry matching. It must recover every minimum restoration set and decide whether a given budget suffices. Neither the visual match nor the graph alone determines the complete answer. A factorial construction independently varies the visual binding and unavailable vertices, with matched answer-preserving visual edits and independent public-ID replicates. The released finite source collection yields 84 constructions from 17 assemblies, grouped into 15 dependence groups, with 1008 repair observations. Exhaustive certificates and an independent semantic scorer verify binding, all optimal solutions, costs and component outcomes. Atomic-binding and oracle-binding conditions separate the two operations. Task-aligned joint outcomes measure multimodal failure on observations solved in both isolated conditions; legal shallow baselines expose avoidable shortcuts. We release construction, exclusion and split ledgers, native inputs, scoring and a versioned analysis path. Learned-model tables remain blank pending collection. The contribution is a controlled test of the binding–optimization interface on this finite resource, not a claim of physical repair or population-wide generalization.

1. Introduction

A vision–language model may identify the right object but fail to reason about the system containing it. It may also solve a graph problem perfectly when the relevant object is named, yet bind that name to the wrong visual candidate. Ordinary task accuracy conflates these failures. The useful question is therefore not merely whether a model can rec-

ognize bricks or solve connectivity problems, but whether the two operations remain correct when their interaction changes under controlled conditions.

BrickAtlas makes this interaction explicit. A supplied undirected graph describes recognized connections inside a real LDraw subassembly. Some vertices are unavailable. Two terminal vertices are named in text; a third is selected by matching an unlabeled reference part to one of four image-labeled candidates. The model must enumerate all smallest restoration sets connecting the three terminals, report their costs and resulting component sizes, and decide whether a stated budget suffices. The geometry match determines which graph problem to solve. The fault state determines how to solve it.

The task has two independently manipulated inputs. A target–distractor label transposition changes the selected terminal while preserving the textual graph and fault set. A different fault assignment changes the optimization problem while preserving the image. Both interventions must change the complete optimal-solution set before a construction is admitted. A distractor–distractor transposition uses the same visual edit mechanism while preserving the answer. Every construction also has independent public-ID assignments, so source numbering cannot serve as an undeclared correspondence rule.

This design supports a specific diagnosis. The atomic condition tests visual binding alone. The oracle condition names the correct terminal and tests structural optimization. The multimodal condition requires both, and the no-image condition retains the full graph without the terminal binding. Pair and factorial scores then test whether correct behavior survives required changes and irrelevant edits. Joint outcomes distinguish failure on individually solved instances from nonoverlapping isolated competence. The task does not measure inference of hidden connectors from pixels: providing the graph deliberately makes structural evidence exact and the remaining visual dependency identifiable.

Our contribution is a source-backed benchmark of this interface, with admissibility based on exact changes to solution sets; matched controls and grouping at construction

time; and semantic scoring that accepts equivalent valid outputs while retaining every failure. The repair objective, CAD assets, shape matching and paired accuracy are established ideas. Their composition is tested here through a fully auditable factorial contract. The benchmark does not assume that learned models outperform shallow systems or that a factorial score must change their ranking.

2. Related Work

Controlled VQA datasets including CLEVR and GQA connect visual questions to explicit scene programs [4, 5]. VQA v2 and contrast sets reduce the value of stable answer priors [2, 3]; consistency tests distinguish invariance from correctness [9]. Winoground and its analysis emphasize that joint failure can mix binding, visibility and off-construct requirements [1, 11]. Our binding task is deliberately narrower than unrestricted referring-expression understanding [8].

VisualFLIP directly evaluates independently queried answer-changing image pairs and competence-conditioned failure [12]. BrickAtlas does not introduce these metrics. Its primary operation is instead a visual-to-graph binding followed by exact optimization, with a second independently varied structural input and a complete solution-set certificate. The distinction concerns inputs and permitted conclusions, not an assertion that other benchmarks lack useful visual controls.

BrickNet provides graph-backed human-authored LDraw assembly programs [6]. LEGO-Puzzles evaluates multistep spatial reasoning [10], while PhyBlock studies physical understanding and assembly planning [7]. Our graph restoration is an abstract decision problem: a restored vertex recovers its original edges. It does not model insertion, extraction, forces or tool use. Table 1 locates this contract among the closest work.

3. Task and Construction

3.1. Public input and semantic output

Let $G = (V, E)$ be the supplied graph, $M \subseteq V$ the unavailable vertices, t_1, t_2 the explicit terminals, and $C \subseteq V$ the four candidate IDs. In image I , $\phi(I, C)$ is the ID of the unique candidate matching the reference geometry, ignoring color and pose. Public IDs are arbitrary references; original instance identities remain private provenance. For a restoration $R \subseteq M$, write

$$G_R = G[V \setminus (M \setminus R)], \quad T = \{t_1, t_2, \phi(I, C)\}.$$

Every restoration costs one per vertex. A feasible R places all vertices in T in one component of G_R . Define

$$k^* = \min_{R \subseteq M: T \text{ connected in } G_R} |R|,$$

$$\mathcal{R}^* = \{R : R \text{ feasible}, |R| = k^*\}.$$

The output includes ϕ , k^* , the budget decision $[k^* \leq b]$, every $R \in \mathcal{R}^*$, and the multiset of component sizes for each corresponding G_R . Order of sets and IDs has no semantic meaning. This richer output prevents credit for one feasible but nonminimal repair or an incomplete list of equally optimal repairs.

The native packet includes the complete question, graph, unavailable IDs, two explicit terminals, candidate IDs, budget and output schema. Only the image supplies the binding. Source names, original IDs, part numbers, coordinates, intervention labels and GT are excluded. The exact per-condition text and PNG hashes define the scored input; human-readable manifests additionally retain grouping metadata.

3.2. Factorial controls and dependency certificates

Each construction has two fault states differing by one unavailable/present exchange, three visual arms, and two independent public-ID replicates. The visual arms are anchor, a matching-candidate label transposition, and a two-distractor label transposition. Thus a construction has twelve repair observations, which remain one dependent unit.

For each fault state, the visual-changing pair must have different complete optimal-solution sets. For the fixed visual binding, the two fault states must also change those sets. The visual-preserving pair must have identical sets, costs and component outputs. All certificates are computed before model evaluation. These are requirements on *semantic* solutions after decoding public IDs, not on changes in JSON order. Figure 2 shows a released construction.

Each candidate card uses actual LDraw geometry at its saved part pose; the reference has a different camera view. Cards are object observations, not proposed freestanding assemblies. Color and numeric ID are not matching criteria. The unedited source assembly is available in the inspection interface to establish provenance; it does not enter the primary request.

3.3. Sampling, exclusions and held-out constructions

The target population is the finite checked-in source collection. We enumerate complete non-root subassemblies, including descendants, with 12–100 parts, a connected recognized-connector graph and full recorded connector coverage. Subassemblies with a recorded unmatched intersection inside the induced set are excluded. This rule is a dataset screen, not a physical certificate.

A seeded ordering selects disjoint instance sets within each source. Aligned typed origin-distance signatures detect repeated designs. Conservative graph-refinement signatures group potential repeated topological templates. Sources linked by contributors, design signatures or topology signatures form transitive dependence groups. Devel-

Visual binding + minimum repair | actual development construction

Match the reference, bind its current graph ID, then return every minimum restoration set.



Structural input

Fixed terminals: V1470, V5885
Unavailable: V5246, V7372, V6716, V2353, V3892, V7052, V4504

Edges: V9801-V8726; V3892-V5885; V9801-V3673; V6716-V7052; V4504-V2353; V6716-V5246; V7052-V3892; V6716-V1470; V6716-V7372; V4504-V2647; V9801-V2353; V6716-V9308; V5246-V3892; V4504-V7372; V6716-V2647; V4504-V8726

GT: terminal V9801; minimum cost 4; budget sufficient: true.

All optimal sets: {V3892, V4504, V5246, V6716} OR {V3892, V4504, V6716, V7052}
Component sizes for each optimum: [11]

Figure 1. An actual primary input and evaluator GT from visual-repair-v1. The supplied graph and unavailable set specify the structural problem; the native image determines the third terminal. Geometry, source poses and labels are taken from the released records. The output requires the bound ID and every minimum restoration, not only the locally matching candidate. GT is shown outside model input. Source context, if shown, is provenance rather than extra input.

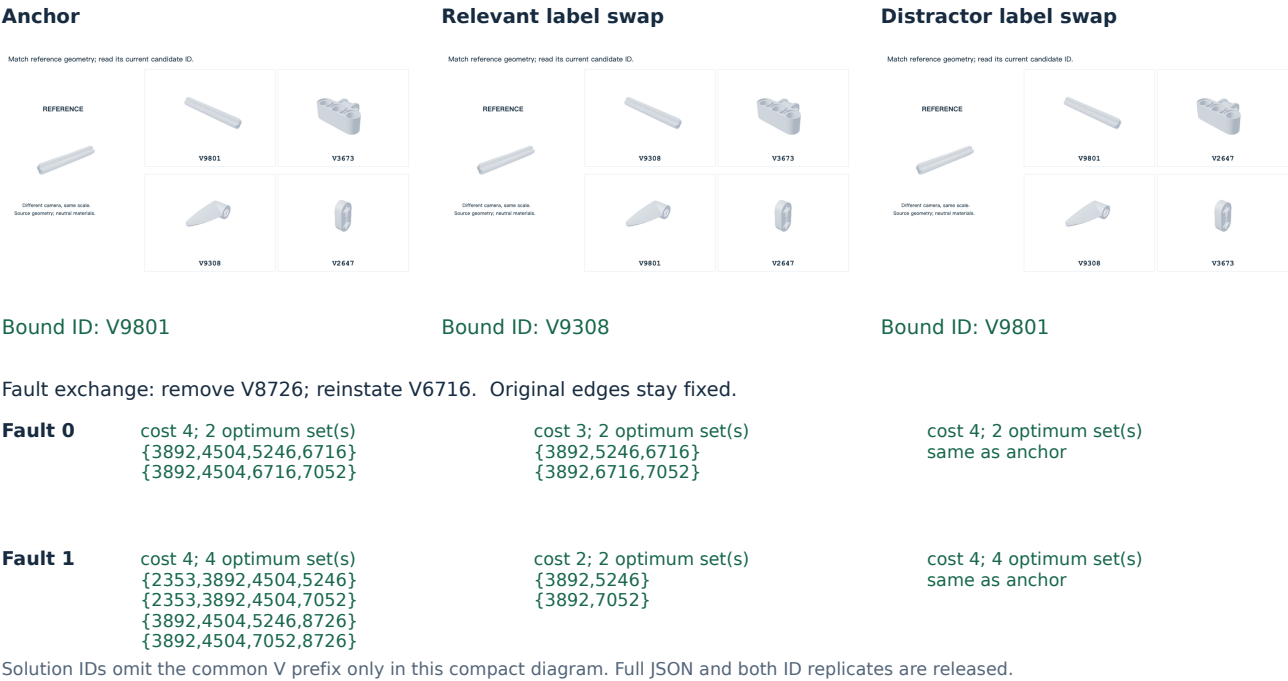
Table 1. Task-level comparison. These are differences in the cited contracts, not claims of performance superiority. BrickAtlas’s certificates concern its supplied graph, not physical feasibility.

Resource	Input and operation	Controlled dependency	Reference answer
VisualFLIP	Image and fixed question	Task-critical image change	Paired question answers
LEGO-Puzzles	Brick scenes and spatial questions	Multistep spatial tasks	Task-specific spatial answers
PhyBlock	Block scenes and assembly tasks	Progressive planning and diagnosis	Planning/physical task criteria
BrickNet	LDraw programs and connector graphs	Generative assembly representation	Source-backed programs
BrickAtlas	Image binding plus supplied graph	Visual × fault change; matched visual preservation	All optimal repair sets, independently checked

opment and held-out groups are assigned before bounded structural search. All views, fault variants and ID replicates of a construction remain together. The ledger retains candidates, exclusions, search attempts and the first admissible result; failed search does not silently disappear from the sampling account.

The final release contains 84 constructions from 17 sources and 15 dependence groups (Table 2). Relevant variation is reported through node/edge ranges, optimal costs and numbers of optimal sets, rather than only source labels. Most observations remain small: 802/1008 have cost two or three, and 598 have one optimum. These counts include dependent variants and are not independent complexity samples. No claim of generalization to all LEGO mechanisms follows from this finite collection. Public CAD reuse may permit contamination; we do not infer confirmed contamination

Visual change, visual preservation and a separate fault exchange



4. Evaluation Protocol

4.1. Semantic correctness and failure handling

The independent scorer uses union-find connectivity rather than the constructor’s graph search and enumerates all candidate restorations. It validates IDs, duplicates, terminal binding, minimum cost, budget decision, complete optimal-set coverage and component results. Alternative valid orderings receive identical credit. Malformed JSON, unknown vertices, omitted optima, extra nonoptimal solutions and inconsistent metadata fail the appropriate criterion. Test fixtures are labeled synthetic and cannot become empirical receipts.

Let $c_{rsav} \in \{0,1\}$ indicate exact correctness for construction r , ID replicate s , structural arm a and visual arm v . Exact correctness requires binding and complete semantic repair. Binding accuracy is also reported independently. The three paired metrics require both members correct: anchor versus visual change, anchor versus visual preservation, and fault zero versus fault one. Factorial correctness is

$$F_{rs} = \prod_{a \in \{0,1\}} \prod_{v \in \{A,C,P\}} c_{rsav}.$$

A constant wrong answer never earns preserving correctness. Conditional repair given correct binding is undefined with zero eligible observations; its denominator accompanies the score.

4.2. Conditions and estimands

The primary multimodal condition solves binding and repair jointly. Atomic binding uses the same image and candidates but omits the graph and fault state. Oracle binding names the true terminal and omits the image. No-image repair includes the graph and candidates without the binding. These conditions distinguish a missing visual dependency from imperfect optimization given a correct binding. They do not prove any particular internal reasoning procedure.

For the same task, ID replicate and intervention cell i , let A_i denote atomic binding correctness, O_i oracle exact repair, M_i multimodal exact repair, and B_i multimodal binding correctness. We retain all eight (A_i, O_i, M_i) contingency counts and define

$$J_{\text{micro}} = \frac{\sum_i A_i O_i (1 - M_i)}{\sum_i A_i O_i}.$$

This is observed joint-condition failure among observations solved in both isolated conditions. It is not a causal measure of an internal reasoning mechanism. A zero denominator is undefined. The primary equal-group J averages the within-group ratios over groups with at least one jointly competent observation; the eligible observation and group counts accompany it. We also report paired differences $M_i - O_i$ in

exact repair and $B_i - A_i$ in binding. Marginal accuracies cannot recover J .

Joins require the same locked model revision, adapter, settings, task membership and qualification eligibility across conditions. Missing or invalid planned calls contribute zero correctness; an unplanned condition makes its contrast unavailable. Micro estimates weight observations; group estimates weight dependence groups equally. Shared-group resampling retains all conditions, images, anchors and ID replicates together. The supplement freezes zero-eligibility and resampling details. Full-set and construction-wide qualified-set outputs retain separate denominators, with pending and rejected counts. Group counts and leave-one-group-out sensitivity accompany results. These are descriptive finite-resource analyses. A worst-case normal half-width target of .20 for independent bounded group means would already require 25 groups before small-sample corrections. More renderings cannot remedy fewer independent groups.

The versioned study manifest binds exact condition inputs, evaluator version, planned slots and baseline configurations. An external run lock must additionally pin provider/model revision, adapter, image handling, output settings and failure policy. Only receipts matching the selected version and input hashes are eligible; a missing slot remains incorrect. A provider-neutral interface does not claim that an unpinned proprietary endpoint is already operational. There are no paid-model results in this release.

5. Planned Experiments

The study asks whether binding-only and oracle-repair accuracy predict success when both are required, whether that success survives the visual and structural interventions, and how much legal shallow or modular systems explain. Null effects and strong baselines are valid outcomes. Table 3 retains the 15-model planning roster: seven proprietary VLMs, six open-weight VLMs and two text-only models. Text-only models are eligible for the symbolic/no-image conditions, not the image-binding condition. Identifiers come from the recorded planning snapshot and require a separate pinned run lock. Unsupported effort-mode comparisons from earlier plans are not part of this version.

The primary result matrix (Table 4) is interpreted together with the atomic, no-image and oracle conditions, reported with the joint-condition matrix in the supplement. A low multimodal score despite strong oracle performance is compatible with binding failure; it is not by itself proof of failed structural reasoning. Likewise, high atomic binding with low oracle repair locates a structural limitation without requiring a visual explanation. Factorial failures can arise at either stage and must be decomposed accordingly. In particular, identical condition accuracies can conceal opposite joint outcomes: J asks whether isolated successes occur on

Table 3. Planned model inventory and input eligibility. These names are planning metadata, not completed runs or a live availability claim. Every unmeasured result is blank.

Model	Model class	Planned inputs
GPT-6 Astra	Proprietary VLM	All four conditions
GPT-5.6 Sol	Proprietary VLM	All four conditions
GPT-5.6 Luna	Proprietary VLM	All four conditions
Claude Fable 5.1	Proprietary VLM	All four conditions
Claude Sonnet 5	Proprietary VLM	All four conditions
Gemini 3.1 Pro Preview	Proprietary VLM	All four conditions
Gemini 3.8 Flash	Proprietary VLM	All four conditions
Qwen3.5-9B	Open-weight VLM	All four conditions
Qwen3.5-27B	Open-weight VLM	All four conditions
InternVL3.5-8B	Open-weight VLM	All four conditions
InternVL3.5-38B	Open-weight VLM	All four conditions
Gemma 3 4B IT	Open-weight VLM	All four conditions
Gemma 3 27B IT	Open-weight VLM	All four conditions
Qwen3-32B	Open-weight text LM	No-image / oracle
DeepSeek-R1-Distill-Qwen-32B	Open-weight text LM	No-image / oracle

Table 4. Primary multimodal results to be collected. Exact requires correct binding and every optimal repair; Bind is terminal accuracy. V-change, V-keep and Fault are both-correct paired metrics. All-six requires every cell of the factorial design correct. All entries are equal-group percentages; micro scores, intervals, failure counts and eligible groups accompany the complete records.

Model	Exact	Bind	V-change	V-keep	Fault	All-six
GPT-6 Astra						
GPT-5.6 Sol						
GPT-5.6 Luna						
Claude Fable 5.1						
Claude Sonnet 5						
Gemini 3.1 Pro Preview						
Gemini 3.8 Flash						
Qwen3.5-9B						
Qwen3.5-27B						
InternVL3.5-8B						
InternVL3.5-38B						
Gemma 3 4B IT						
Gemma 3 27B IT						

the same observations as multimodal failures. Zero J , either sign of a paired difference, and no eligible observations are all permitted outcomes.

All response slots and revisions must be frozen before real collection. The current release provides deterministic construction and baseline audits plus executable scoring; planned model and human results are not represented by fabricated values. Historical Color, Part-type and small graph studies remain separately versioned and are not pooled with the new repair family.

6. Limitations

BrickAtlas isolates one visual-to-symbolic interface. It does not recover connectivity from images, model the feasibility

of inserting a part, certify stable physical assemblies, or evaluate unrestricted repair planning. Candidate cards make the binding problem explicit and may favor shallow appearance matching; such performance narrows what end-to-end errors can establish. Recognized connectors may be incomplete despite exact graph evaluation. Geometry equality does not guarantee human-visible uniqueness, making native-input qualification necessary.

The finite collection has dependent sources, repeated part types and limited construction groups. Grouping is conservative, not proof that all latent similarities were found. Held-out group performance supports claims within the declared construction population only. Selection requires informative counterfactuals, so the benchmark does not estimate natural failure frequency.

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